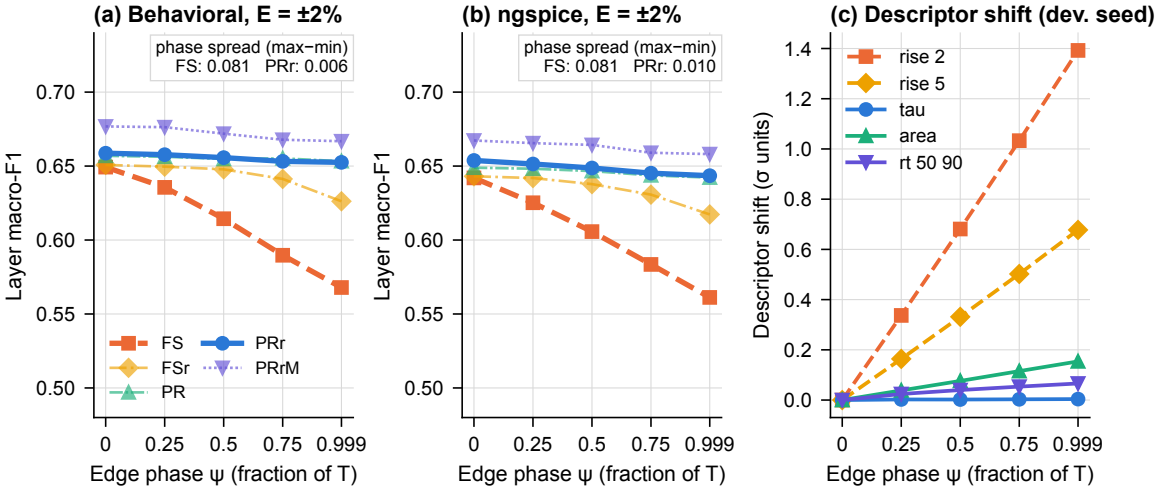
(a) Behavioral, E = ±2%
phase spread (max−min)
FS: 0.081 PRr: 0.006
Layer macro-F1
0.70
0.65
0.60
0.55
0.50
FS
FSr
PR
PRr
PRrM
0
0.25
0.5
0.75
0.999
Edge phase ψ (fraction of T)
(b) ngspice, E = ±2%
phase spread (max−min)
FS: 0.081 PRr: 0.010
Layer macro-F1
0.70
0.65
0.60
0.55
0.50
0
0.25
0.5
0.75
0.999
Edge phase ψ (fraction of T)
(c) Descriptor shift (dev. seed)
Descriptor shift (σ units)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
rise 2
rise 5
tau
area
rt 50 90
0
0.25
0.5
0.75
0.999
Edge phase ψ (fraction of T)